

Question:

Which one of the following statements is NOT CORRECT?

- 1) A well-conditioned matrix has a condition number close to 1
- 2) An ill-conditioned matrix has a large condition number
- 3) The inverse of a well-conditioned matrix can be computed accurately
- 4) A non-invertible matrix has a condition number close to 1

Solution:

The **condition number** of an invertible matrix $\mathbf{A}$, denoted as $\kappa(\mathbf{A})$, is defined as:

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\| \quad (1)$$

It measures how much the output of a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ can change for a small change in the input data.

- 1) **A well-conditioned matrix has a condition number close to 1:** This is **CORRECT**. The smallest possible condition number is 1 (for example, the identity matrix). A value close to 1 indicates that the matrix is numerically stable.
- 2) **An ill-conditioned matrix has a large condition number:** This is **CORRECT**. This is the definition of an ill-conditioned matrix. A large condition number means that small errors in the input can lead to very large errors in the solution.
- 3) **The inverse of a well-conditioned matrix can be computed accurately:** This is **CORRECT**. Because a well-conditioned matrix is not sensitive to small perturbations, numerical algorithms used to compute its inverse will produce results with high accuracy.
- 4) **A non-invertible matrix has a condition number close to 1:** This is **NOT CORRECT**. A non-invertible (or singular) matrix does not have an inverse $\mathbf{A}^{-1}$. For the purposes of conditioning, its inverse is considered to have an infinite norm. Therefore, the condition number of a non-invertible matrix is defined to be **infinity** (∞). An infinite condition number is the largest possible value, representing the worst possible conditioning, not the best (which is close to 1).